Process to screen soil sample datasets
Supporting Document `IndigoCarbon_US-1_2023_0060`

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1 Context

Following sample collection and subsequent lab analysis, the soils dataset undergoes an internal QC process to ensure sample integrity and final preparation for use as a model input.

1.1 Definitions

- **Call Plan:** The highest level of data collection form organization in Indigo Fields that can be applied to a single field for a specific soil sampling campaign. For example, soil sampling conducted in the fall 2020 sampling season was collected in the Carbon Soil Sampling v2 call plan. Samples collected in the same field, but as part of different sampling campaigns, may be entered into separate call plans.

1.2 Initial Valid, Unique Sample Data Screening

The first stage of the Indigo Ag's soil data QC process ensures that the soil sample data entered into Indigo Fields is unique to a single specific sample in the program and can be reliably paired with the analysis results being sent from the contracted soil labs. To pass the first step, both the Indigo Fields entered sample data and soil analysis results data must be deemed valid, unique, and not deleted according to the following criteria:

- **Valid:** All QR code entries in Indigo Fields and soil lab results returned must be considered "valid." A valid QR code is 10-15 characters in length and contains no spaces.
  - NOTE: A small number of QR codes from the beginning of the program may be 9 characters and are handled as exceptions on a case-by-case basis
- **Unique:** All soil sample QR codes entered into Indigo Fields Mobile and returned as soil lab results are required to be unique in the Indigo database. A QR code is considered unique if it appears only once as an Indigo Fields entry and once as a soil lab result. This ensures that the correct lab result is paired with the correct sampling information
- **Not Deleted:** For a variety of reasons, such as improper sample handling in the field, accidental destruction in the soil lab, and an inability to unambiguously identify a sample, an Indigo Fields sample record and/or

soil analysis result may be "soft deleted." This process allows for the sample records to be retained, but not included in the creditable data set. To pass stage one, a sample must not be soft deleted for any reason.

If a QR code is invalid and/or not unique, QR code entries in Indigo Fields and/or results from the soil analysis labs may be modified to establish validity or uniqueness should the correction not compromise the integrity of the sampling data. If the integrity of the sample cannot be guaranteed, the sample information is deleted to be removed from program carbon quantification, however the original entry is retained in the Indigo database to maintain the record for internal use.

If the sample Indigo Fields entry and result are valid, unique, and not deleted, the two are paired and moved to the next step in the process.

1.3 Calculations

Once the data collected in Indigo Fields is reliably paired, the data may undergo one or more of the following calculations to prepare for the final stage of QC.

Censored Values Some analysis labs are uncomfortable reporting organic carbon concentration measurements below 0.20% due to methodological limits of detection. In instances where the measurement is returned as '< 0.20', the value is converted to 0.10% and labelled as a censored value. Because we cannot know the true value of the measurement, the midpoint of possible outcomes is selected as it represents an equal probability that the true value is over- or underestimated in each individual circumstance. If a lab reports a real value below 0.20%, the real, reported value is used.

Sampling Depth If sample depth is not recorded at the time of Indigo Fields entry, because the sample was collected to full depth according to `IndigoCarbon_US-1_2023_0001` and `IndigoCarbon_US-1_2023_0002` or if the sample depth is erroneously recorded as zero yet a sample is submitted to the lab, the depths are assumed as:

- Soil Organic Carbon – 30 cm
- Bulk Density – 30.48 cm

If a sample depth entry is recorded in Indigo Fields, the sample depth achieved is reported in centimeters.

Bulk Density Bulk density is calculated based on the fine earth fraction mass returned by the soil analysis labs:

- BD (g/cm^3) = fine earth fraction mass in g / (3.14 x (2.54 cm^2) x sample depth in cm)
- Therefore, for samples where the full 30.48 cm depth was reached:
  - BD (g/cm^3) = fine earth fraction mass in g / (617.46 cm^3)
- NOTE: Coarse material is not included in the bulk density calculation as this fraction is not known to contain significant amounts of carbon, nor is this fraction represented in the carbon analysis.

Distance Between Actual Sampling Location and the Prescribed Location The sampling data entered according to IndigoCarbon_US-1_2023_0004 into the IF Soil Sampling form allows for the comparison of the location for which a sample is prescribed relative to the location in which the sampler records the sampling location. This allows Indigo to ensure that samples are being collected according to protocol specifications.

This calculation in feet is performed using the SQL functions st_distance() and st_makepoint() along with the GPS coordinates of the appropriate records.

Distance Between Actual Sampling Location and the Physical Location of the Sampler The sampling data entered according to IndigoCarbon_US-1_2023_0004 into the Indigo Fields Soil Sampling form allows for the comparison of the location where the sampler is entering data relative to the location in which the sampler records the actual sampling location. This allows Indigo to ensure that samplers in the field are sampling where they claim to be.

This calculation in feet is performed using the SQL functions st_distance() and st_makepoint() along with the GPS coordinates of the appropriate records.

1.4 Final QC Check

Following successful completion of the initial QC screening and undergoing the calculations outlined above, soil samples must pass all the following checks to be included in the creditable soil dataset.

As prescribed This check ensures that the sample collected at the designated point was the type of sample that was prescribed for that location.

- **Prior to September 1, 2020:** Because the samples were not directly prescribed in Indigo Fields, this check is considered "passed" for this timeframe if the actual sample point location is correctly recorded.
- **September 1, 2020 to Present:** To pass, the sample type collected is required to be prescribed at the location where it was entered.
  - NOTE: On the rare occasion that a bulk density sample is collected at a location not prescribed, this QC flag may be overwritten and documented allowing the sample into the dataset based on the rationale that more bulk density measurements are likely to improve carbon stock estimates.
  - This parameter does not apply to the Texture/pH samples given that they are collected at the field-level, not point-level.

Correct Analysis This parameter ensures that the lab result returned is the lab result expected based on the Indigo Fields entry. For each sample type Indigo expects the following results:

- Soil Organic Carbon: Excess lime determination and appropriate carbon measurement via dry combustion analysis
- Bulk Density: dry mass of the fine earth (< 2mm) fraction
- pH: 1:1 soil:water pH value
- Texture: percent by mass of the clay, silt, and sand fractions

Samples receiving the wrong analysis are excluded from the program.

Point moved beyond SOP limit The protocol to select alternative sample locations is outlined in IndigoCarbon_US-1_2023_0004. This parameter checks that samplers have not move points in the field more than 10 ft beyond the SOP limit. If the actual sampling distance relative to the prescribed location recorded by the sampler is farther than 10 ft beyond the SOP limit, the sample is excluded from the program.

- NOTE: a 10 ft buffer was established to allow some tolerance for the combined error of GPS accuracy and placing the location pin on the IF map when a sampler is properly executing the protocol in the field.

Allowable Distance (including 10 ft buffer):

- October 1, 2019 – February 28, 2021: $\leq$ 60 ft
- March 1, 2021 – Present: $\leq$ 35 ft

Sample Record Location Corresponds to Sampling Location Record Sampling responsibilities are conducted in remote areas afield, so it is important for Indigo to confirm that the samples are being collected where they are prescribed and recorded to be by samplers. To do this, Indigo Fields records the GPS location of the device running Indigo Fields when a sampling location is recorded.

Because samples being recorded away from the exact location of collection does not immediately mean the collection is in error, a two-stage review is used to provide a reasonable level of assurance that the sampling protocol is being followed correctly (Figure 1). Reasons device location may deviate from the recorded sampling location include but are not limited to data entry correction, record of information at an RTV parked central to several prescribed locations, and failure of the sampler to retain a supplemental GPS receiver on their person at the time of data entry (e.g. the receiver is placed in the cupholder of an RTV parked some distance from the sampling point while the device recording information is taken to the location of sampling thus erroneously logging the incorrect location).

Stage One – Sampler-Level: An initial screening threshold of 100 ft was established to indicate if the recording device is located at a substantial distance from the recorded sampling location for every sample that a soil sampler collects.

- **0 - 50% of samples recorded by sampler exceed the initial screening threshold (100 ft):** Samples are subject to "Stage Two" scrutiny below.
- **> 50% of samples recorded by sampler exceed the initial screening threshold (100 ft): ALL** samples collected by the sampler (including those within the threshold limit) are removed from the dataset unless a sampler can produce an explanation considered sufficient to explain the observed behavior.

Stage Two – Field-Level:

- **0-5% of samples collected in a field exceed the initial screening threshold (100 ft):** No further action is taken.
- **> 5% of samples collected in a field exceed the initial screening threshold (100 ft):**
  - As of 2023, data from previous visual review processes were used to construct a set of automated checks that find the most common cases for fields failing the initial screening. Fields that pass this automated review have no further action taken and fields that fail are visually reviewed for data integrity.
  - Visually compare sampling patterns for consistency and systematic bias.

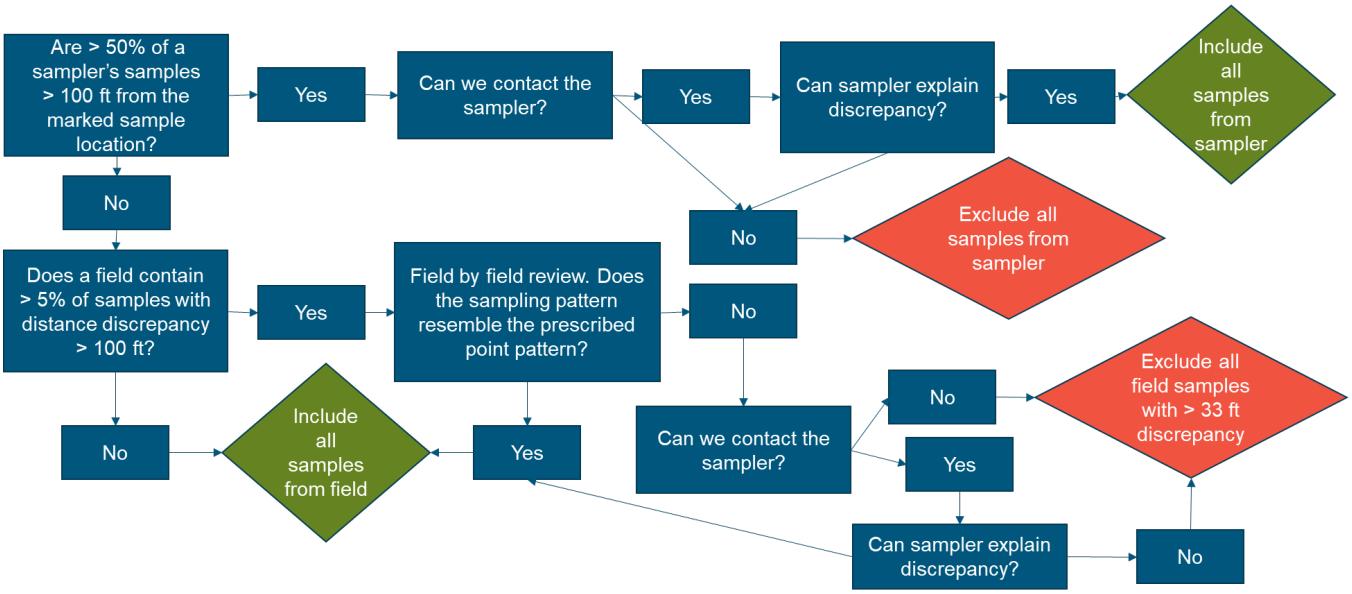


Figure 1: Indigo Fields flow logic

Unless a sampler can produce an explanation considered sufficient to explain the observed behavior, all samples recorded > 33 ft from the actual sample location are excluded from the dataset.

Inadequate Sample Depth Samples collected to a depth of 23 cm and greater were used in the calibration dataset for the DayCent-CR model. To best align with the calibration approach, the cutoff for the rare samples collected to less than full depth was assigned at 23 cm. Soil carbon and bulk density samples collected at depths less than 23 cm are excluded from the program.

2 Sample Shipment Timeframe (Applies March 1, 2021 – Present)

Improvements in the sample monitoring process have allowed for Indigo to receive confirmation of when samples are received by the contacted soil labs. SOPs outlined in IndigoCarbon_US-1_2023_0006 require samples to be shipped no more than 4 days after collection. Because of Indigo's limited control of the shipment process once leaving

the samplers control, a tolerance of 21 days from the date of collection has been set for samples to reach the lab. Samples reaching the soil analysis lab <= 21 days from the date of collection, based on the date the sample is entered into Indigo Fields, are included in the dataset. If the date of sample receipt by the lab is for some reason unknown, samples where the lab testing result is received by Indigo <= 49 days from the date in which the sample was entered into Indigo Fields are considered acceptable. This encompasses the original 21-day tolerance and an additional 28 days given to the lab to complete analysis and report results based upon contractual obligations.

2.1 Location is sampled twice

Carbon and Bulk Density Samples If, on the rare occasion, more than one carbon or bulk density sample is collected on the exact same location (i.e., multiple samples, same point id), one sample per sample type is selected to represent the point according to the following hierarchy.

For example, if two samples that passed all QC parameters were collected on the sample point and one sample was collected in frozen soil, while the other was collected in soil that was not frozen, the sample collected in soil that was not frozen would be selected to represent the point.

1. Passed all sample QC checks outlined above
2. Soil was not frozen when collected
3. Sample with deeper sampling depth
    - i.e., for carbon, sample with sampling depth closer to 30 cm (SOC), 30.48 cm (Bulk Density)
4. Sample point was not offset (an alternative location to the prescribed location was not selected.
    - If both samples are offset, the sample with the actual sample location closer to prescribed sample location is chosen
5. Sample date closer to the mode sample date on the field (i.e., when the most other samples were collected on the field). If the date difference is the same, take the later date.
6. If, by any chance, the above are all the same, rank the samples by QR code in ascending order and select the first

Texture/pH Composite Sample If, on the rare occasion, more than one composite sample was collected on the same field/call plan the following process converts the samples into a single value representing the field according to the process below.

1. Take the average of each measurement recorded within the same call plan. This process is used to simulate the compositing process outlined in IndigoCarbon_US-1_2023_0003.
    a. Multiple texture/pH samples within the same call plan implies that the texture/pH samples may not be composite samples. Taking the average converts the measurements into a pseudo-composite (i.e., averaging the results of four individual samples is approximately equivalent to a single result from a homogenized sample in which each of the four individual samples makes an equal contribution).
2. Then take the average within the same sample design across call plans.
    a. Thus, get the average measurements per (field, carbon sample design)

    NOTE: Average pH is calculated by:

1. Converting the pH value of each reported result to the hydronium concentration
2. Averaging the appropriate hydronium concentrations together
3. Converting the average hydronium concentration back to pH

    Because the reported value "pH" is the negative logarithm base 10 of the hydronium concentration, the concentrations must be used when averaging. Averaging the pH values themselves would result in an erroneous mean value.

For example, the average of pH 5 and pH 6 equals 5.3, not 5.5 as would be the result of average "5" and "6." This is because the hydronium concentration of a sample at pH 5 is ten times the concentration of a sample at pH 6.

NOTE: Texture results from the averaging of each individual attribute (e.g. percentage clay, percentage silt, percentage sand).

Outliers Detection and Handling (Ranges updated as of May 1, 2023)

Soil Carbon: Soil organic carbon data distribution is allocated in the following classes:

- 0 to <= 7% = High confidence modeling range
- >7% = lower confidence modeling range

Bulk density: Bulk density data distribution is allocated in the following classes:

- > 0 g/cm3 to < 0.7 g/cm3: Low bulk density
- >= 0.7 g/cm3 to <= 2.2 g/cm3: Agronomically acceptable bulk density
- > 2.2 g/cm3: High bulk density

To ensure that modeled carbon sequestration is of the highest quality, outlier handling has been streamlined to restrict soil organic carbon values from the creditable dataset. The above distributions resulted from an analysis to determine a range of soil organic carbon values that can reasonably be simulated using the DayCent model. Therefore, soil organic carbon values above 7% are restricted from modeling. Bulk density samples that are considered outside of an agronomically acceptable range are restricted from the creditable dataset as well. Where there are valid soil organic carbon values but out of range bulk density values, the value is imputed using an algorithm that predicts bulk density given nearby measurements and soil survey measurements.

3 Summary of Changes

Section	Date	Summary of Changes
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